

# Chapter 3 – Vectors

Coordinate systems  
Properties of vectors  
Unit vectors

Vectors and scalars  
Components  
Multiplying vectors

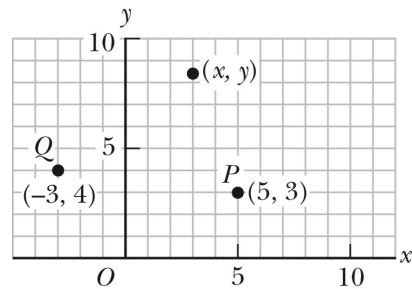
## Coordinate System

- ▶ Physics often requires a description of a location in space
  - e.g. an objects motion as in previous chapter 2.
- ▶ Use various ways to represent this location
  - Cartesian or Rectangular coordinates
  - Plane polar coordinates
- ▶ Need to apply trigonometric functions

## Cartesian or Rectangular Coordinate System

$x$ - and  $y$ - axes intersect at the origin

Points are labeled  $(x, y)$



3

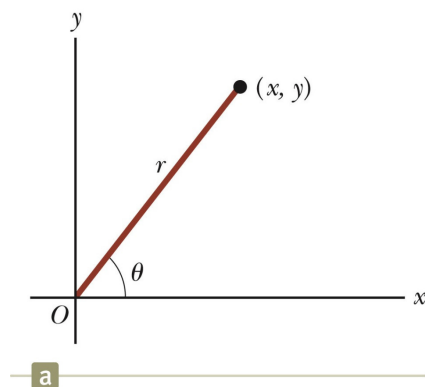
## Polar Coordinate System

Origin and reference line are noted

Point is distance  $r$  from the origin in the direction of angle  $\theta$ , counter clockwise from reference line

- The reference line is often the  $x$ -axis.

Points are labeled  $(r, \theta)$



4

## Polar ↔ Cartesian Coordinates

Based on forming a right triangle from  $r$  and  $\theta$

$$x = r \cos \theta$$

$$y = r \sin \theta$$

If the Cartesian coordinates are known:

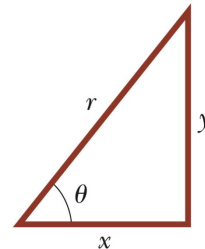
$$\tan \theta = \frac{\text{opp}}{\text{adj}}$$

$$r = \sqrt{x^2 + y^2}$$

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

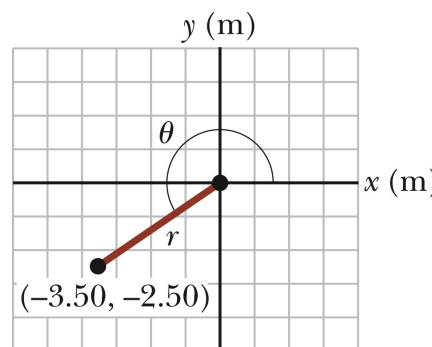
$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$



5

## Example 3.1

- ▶ Work through
- ▶ The Cartesian coordinates of a point in the  $xy$  plane are  $(x, y) = (-3.50, -2.50)$  m, as shown in the figure. Find the polar coordinates of this point.
- ▶ Repeat with the reference line along the  $-x$ -axis



6

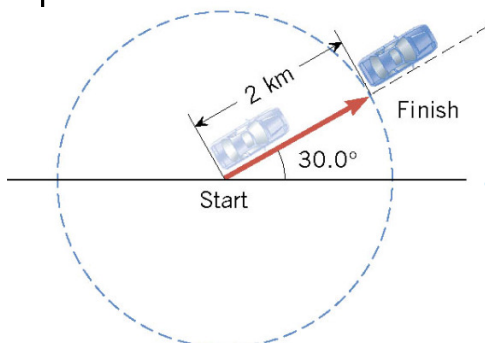
# Scalars and Vectors

- ▶ A **scalar** quantity is completely described by a single value with an appropriate unit and has no direction.
  - temperature, speed, mass
- ▶ A **vector** is a physical quantity and is completely specified by a number with an appropriate unit (magnitude) plus a direction:
  - velocity, force, displacement
  - $\vec{F}$  or  $\vec{F}$  or  $F$  or  $\vec{F}$  in texts
  - Handwritten:  $\vec{F}$
  - Magnitude of the vector:  $|\vec{F}|$

$$\therefore F = |\vec{F}|$$

7

# Scalars and Vectors

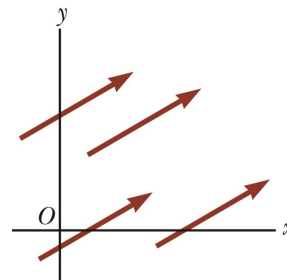
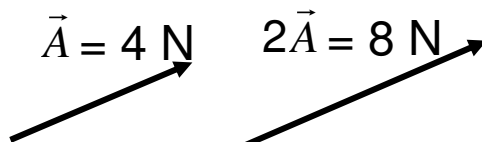
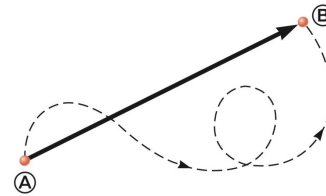


- ▶ The car moved a distance of 2 km.
- The car moved a distance of 2 km in a direction  $30^\circ$  north of east.

8

# Vectors

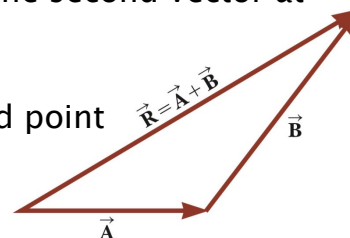
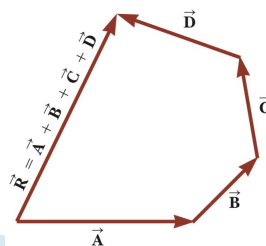
- ▶ Arrows are used to represent vectors. The direction of the arrow gives the direction of the vector.
- ▶ By convention, the length of a vector arrow is proportional to the magnitude of the vector.



9

## Vector addition–Geometric

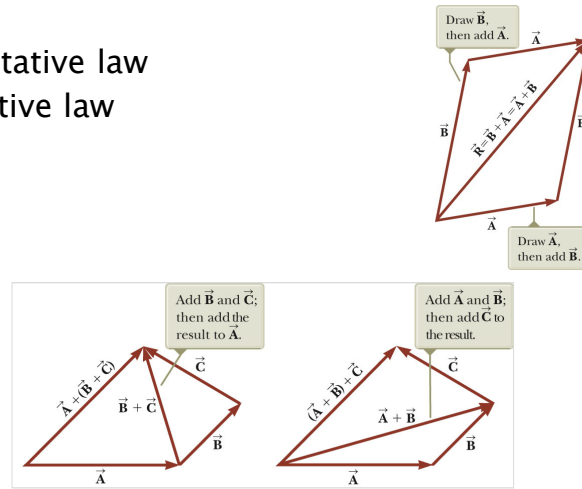
- ▶ A particle moves from point 1 to point 2, then from point 2 to point 3.
- ▶ Draw first vector to scale, then the second vector at the tip of first.
- ▶ “Head to tail” method
- ▶ Resultant = Starting point to End point



10

## Vector addition–Geometric

- ▶ Commutative law
- ▶ Associative law

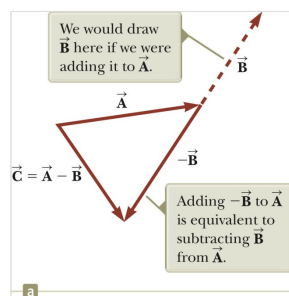


$$\vec{A} + (\vec{B} + \vec{C}) = (\vec{A} + \vec{B}) + \vec{C}$$

11

## Vector addition–Geometric

- ▶ Vector subtraction  $\vec{A} + (-\vec{B}) = \vec{C} = \vec{A} - \vec{B}$   
(Negative vector)



12

## Example

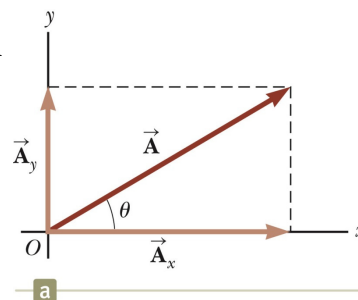
- ▶ The magnitudes of displacements  $\vec{a}$  and  $\vec{b}$  are 3m and 4 m respectively, and  $\vec{c} = \vec{a} + \vec{b}$ . Considering the various orientations of  $\vec{a}$  and  $\vec{b}$ , what is the (i) maximum and (ii) minimum possible magnitude for  $\vec{c}$ ?

Also go through Example 3.2

13

## Components of vectors

- ▶ Graphical addition of vectors – inefficient
- ▶ A component of a vector is a projection of the vector on an axis.
- ▶  $\vec{A}_x$  is the component of vector  $\vec{A}$  along the  $x$ -axis, and  $\vec{A}_y$  is the component of vector along the  $y$ -axis.
- ▶ Draw  $\perp$  lines to axes.
- ▶ Resolve the vector.
- ▶  $A_x, A_y$  both positive in this case.
- ▶ Generally, 3 components  $x$ -,  $y$ - and  $z$ -components.



14

# Components of vectors

► Usually:

$$\cos \theta = \frac{\text{adj}}{\text{hyp}}$$

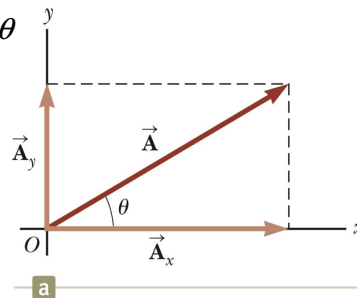
$$\text{then } A_x = A \cos \theta$$

$$\sin \theta = \frac{\text{opp}}{\text{hyp}}$$

$$\text{then } A_y = A \sin \theta$$

►  $A = \sqrt{A_x^2 + A_y^2}$

$$\tan \theta = \frac{\text{opp}}{\text{adj}} = \frac{A_y}{A_x}$$



15

# Components of vectors

► Now if you use the other angle –  $\phi$ :

$$\cos \phi = \frac{\text{adj}}{\text{hyp}}$$

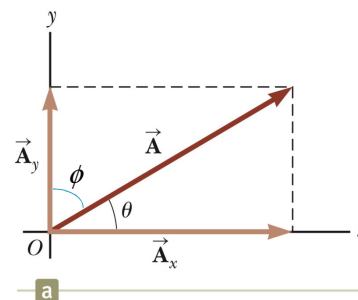
$$\text{then } A_y = A \cos \phi$$

$$\sin \phi = \frac{\text{opp}}{\text{hyp}}$$

$$\text{then } A_x = A \sin \phi$$

►  $A = \sqrt{A_x^2 + A_y^2}$

$$\tan \phi = \frac{A_x}{A_y}$$



16



## Example

- ▶ A small plane leaves an airport on an overcast day and is later sighted 215 km away in a direction making an angle of  $22^\circ$  east of due north. How far east and north is the plane from the airport when sighted?

17

## Example

- ▶ For 2 decades spelunking teams sought a connection between Flint Ridge cave system and Mammoth Cave (Kentucky, USA) When the connection was finally discovered, the combined system was declared the world's longest cave ( $>200\text{km}$ ) The team that found the connection had to crawl, climb and squirm through many passages, travelling a net 2.6 km westward, 3.9 km southward and 25m upward. What was their displacement from start to finish?

18

# Unit Vectors

- ▶ A unit vector has a magnitude of exactly 1 and points in a particular direction.
- ▶ Dimensionless and unitless
- ▶ Purpose is to specify direction
- ▶ Labelled:  $\hat{i}$ ,  $\hat{j}$  and  $\hat{k}$  for  $x$ ,  $y$ ,  $z$  direction

- Right handed co-ordinate system.

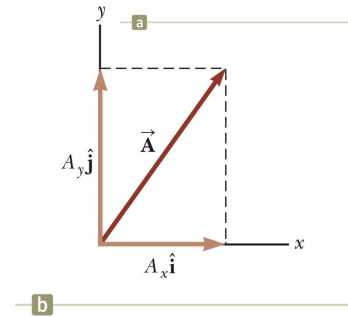
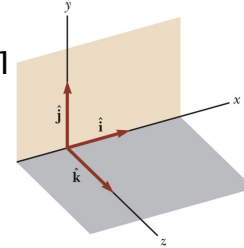
$$\vec{A} = A_x \hat{i} + A_y \hat{j}$$

- Vector components (units)

$$\vec{A}_x = A_x \hat{i} \text{ and } \vec{A}_y = A_y \hat{j}$$

- Scalar components (units)

$$A_x \text{ and } A_y$$



19

# Adding Vectors by Components

- ▶ Consider the vectors

$$\vec{a} = a_x \hat{i} + a_y \hat{j} + a_z \hat{k}$$

$$\vec{b} = b_x \hat{i} + b_y \hat{j} + b_z \hat{k}$$

$$\vec{r} = \vec{a} + \vec{b}$$

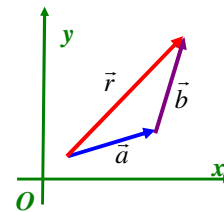
- ▶ Then

$$r_x = a_x + b_x$$

$$r_y = a_y + b_y$$

$$r_z = a_z + b_z$$

$$\vec{r} = r_x \hat{i} + r_y \hat{j} + r_z \hat{k}$$



20

# Subtraction of Vectors

- Consider the vectors

$$\vec{a} = a_x \hat{i} + a_y \hat{j} + a_z \hat{k}$$

$$\vec{b} = b_x \hat{i} + b_y \hat{j} + b_z \hat{k}$$

$$\vec{d} = \vec{a} - \vec{b}$$

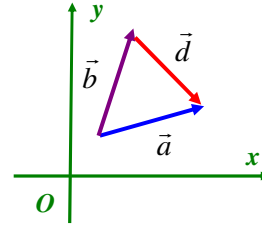
- Then  $d_x = a_x - b_x$

$$d_y = a_y - b_y$$

$$d_z = a_z - b_z$$

- and

$$\vec{d} = d_x \hat{i} + d_y \hat{j} + d_z \hat{k}$$



21

## Example

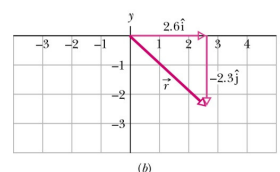
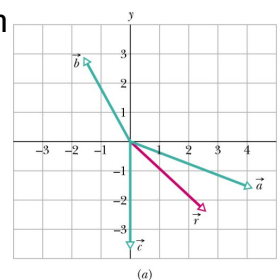
- Consider the three vectors as shown in the figure.

$$\vec{a} = (4.2 \text{ m})\hat{i} - (1.5 \text{ m})\hat{j}$$

$$\vec{b} = (-1.6 \text{ m})\hat{i} + (2.9 \text{ m})\hat{j}$$

$$\vec{c} = (-3.7 \text{ m})\hat{j}$$

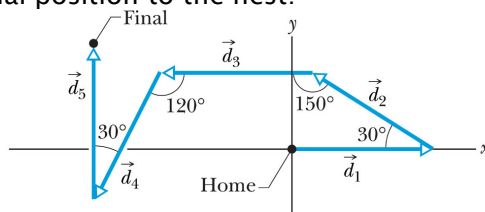
What is their vector sum (also shown)?



22

## Example

- ▶ When an ant wants to return to its home next, it effectively sums its displacements along the axes of the system to calculate a vector that points directly home. Consider an ant making five runs of 6.0 cm each as shown in figure.
  - What are the magnitude and direction of the ant's net displacement vector?
  - What are the magnitude and direction of the homeward vector from the final position to the nest?



23

## Multiplying Vectors

- ▶ Multiplying a vector by a scalar
  - new vector  $s$  (magnitude of vector)
  - Direction is same as original vector if  $s$  is positive
  - Divide vector by  $s =$  multiply by  $1/s$ .
- ▶ Multiplying a vector by a vector
  - Two methods
  - One method produces a scalar – **scalar product**
  - One method produces another vector – **vector product**

24

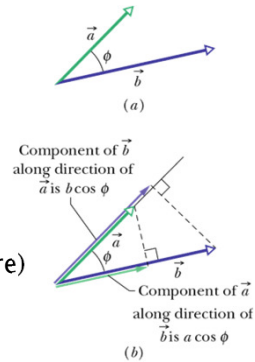
# Scalar product

- ▶ Two vectors  $\vec{a}$  and  $\vec{b}$
- ▶ Scalar product or dot product
 

$$\vec{a} \cdot \vec{b} = ab \cos \phi$$

  - $a, b$  magnitudes of the vectors
  - $\phi$  - angle between the directions of two vectors
- ▶ The product of the magnitude of one vector and the scalar component of the other vector along the direction of the first. (see (b) in figure)
- ▶ If  $\phi = 0^\circ$  then dot product is a max.
- ▶ If  $\phi = 90^\circ$  then dot product is 0.
- ▶  $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$
- ▶ 

$$\vec{a} \cdot \vec{b} = a_x b_x + a_y b_y + a_z b_z$$



25

# Scalar Product

- ▶ Scalar product using unit vector notation

$$\begin{aligned}
 \vec{a} \cdot \vec{b} &= (a_x \hat{i} + a_y \hat{j} + a_z \hat{k}) \cdot (b_x \hat{i} + b_y \hat{j} + b_z \hat{k}) \\
 &= (a_x \hat{i} \cdot b_x \hat{i}) + (a_x \hat{i} \cdot b_y \hat{j}) + (a_x \hat{i} \cdot b_z \hat{k}) \\
 &\quad + (a_y \hat{j} \cdot b_x \hat{i}) + (a_y \hat{j} \cdot b_y \hat{j}) + (a_y \hat{j} \cdot b_z \hat{k}) \\
 &\quad + (a_z \hat{k} \cdot b_x \hat{i}) + (a_z \hat{k} \cdot b_y \hat{j}) + (a_z \hat{k} \cdot b_z \hat{k}) \\
 &= a_x b_x + a_y b_y + a_z b_z
 \end{aligned}$$

26

## Example

- ▶ Vectors  $\vec{C}$  and  $\vec{D}$  have magnitudes of 3 units and 4 units, respectively. What is the angle between the directions of the vectors if the dot product equals (i) 0, (ii) 12 units, and (iii)  $-12$  units?

27

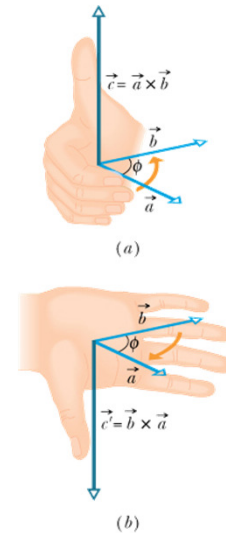
## Examples

- ▶ What is the angle  $\phi$  between  $\vec{a} = 3.0\hat{i} - 4.0\hat{j}$  and  $\vec{b} = -2.0\hat{i} + 3.0\hat{k}$  ?

28

# Vector Product

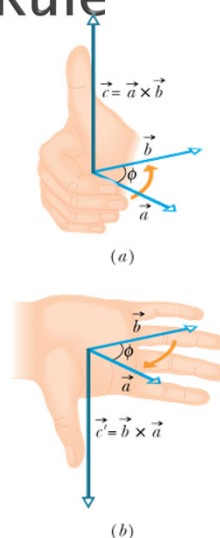
- ▶ Two vectors  $\vec{a}$  and  $\vec{b}$
- ▶ Vector product or cross product
  - $c = |\vec{a} \times \vec{b}| = ab \sin \phi$
  - $c$  - Magnitude of the third vector
  - $a, b$  magnitudes of two initial vectors
  - $\phi$  - smaller of the two angle between the two vectors.
- ▶ If  $\phi = 0^\circ$  then cross product is 0.
- ▶ If  $\phi = 90^\circ$  then cross product is a max.
- ▶ Direction of  $\vec{c}$  is  $\perp$  to plane containing  $\vec{a}$  and  $\vec{b}$
- ▶ **RIGHT-hand rule:**



29

# Vector Product – RH Rule

- ▶  $\vec{c} = \vec{a} \times \vec{b}$
- ▶ Place the vectors tail to tail.
- ▶ Point the fingers of your right hand in the direction of the first vector.
- ▶ your fingers, through the smallest angle between  $\vec{a}$  and  $\vec{b}$ , from vector  $\vec{a}$  to  $\vec{b}$ .
- ▶ Your outstretched thumb points in the direction of  $\vec{c}$ .
- ▶ Commutative law does NOT apply.
 
$$\vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$$



30

# Vector Product

- ▶  $\vec{c} = \vec{a} \times \vec{b}$
- ▶  $\vec{a} \times \vec{b} = (a_x \hat{i} + a_y \hat{j} + a_z \hat{k}) \times (b_x \hat{i} + b_y \hat{j} + b_z \hat{k})$ 

$$= (a_x \hat{i} \times b_x \hat{i}) + (a_x \hat{i} \times b_y \hat{j}) + (a_x \hat{i} \times b_z \hat{k})$$

$$+ (a_y \hat{j} \times b_x \hat{i}) + (a_y \hat{j} \times b_y \hat{j}) + (a_y \hat{j} \times b_z \hat{k})$$

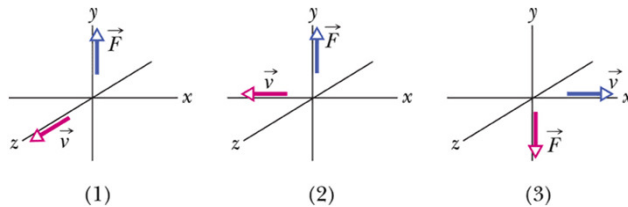
$$+ (a_z \hat{k} \times b_x \hat{i}) + (a_z \hat{k} \times b_y \hat{j}) + (a_z \hat{k} \times b_z \hat{k})$$

$$= (a_y b_z - b_y a_z) \hat{i} + (a_z b_x - b_z a_x) \hat{j} + (a_x b_y - b_x a_y) \hat{k}$$
- ▶ Determinant method

31

## Example

- ▶ If  $\vec{F} = q(\vec{v} \times \vec{B})$  and  $\vec{v}$  is perpendicular to  $\vec{B}$ , then what is the direction of  $\vec{B}$  in each of the three situations below:

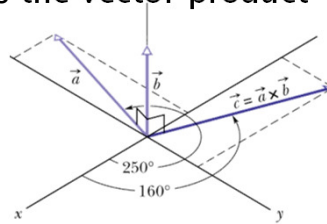


32



## Example

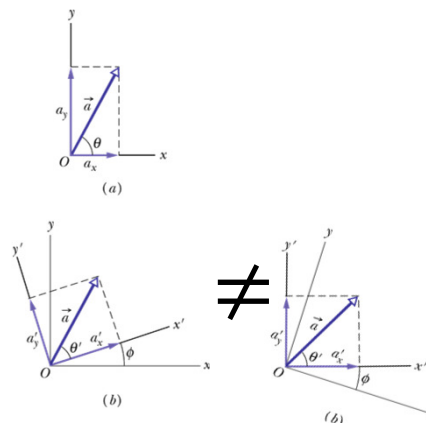
- ▶ The figure shows vector  $\vec{a}$  lying in the  $xy$  plane, with a magnitude of 18 units and pointing in the direction of  $250^\circ$  from the  $+x$ -direction. Also, vector  $\vec{b}$  has a magnitude of 12 units and points along the  $+z$  direction. What is the vector product  $\vec{c} = \vec{a} \times \vec{b}$ ?



33

## Vectors and Physics

- ▶ Usually have shown axes for vectors parallel to the page edges.
- ▶ No particular reason for this.
- ▶ Can rotate the axis and still have the "same" resultant vector
- ▶ Use this a lot in Physics particularly Mechanics.
- ▶ You CANNOT rotate the vectors!!



34